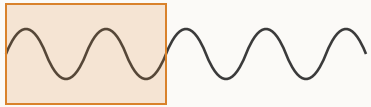
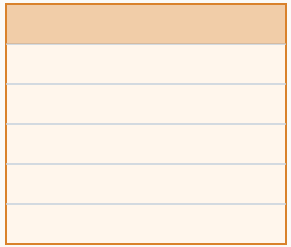


overlapping windows, hop 256



frames F × 1024

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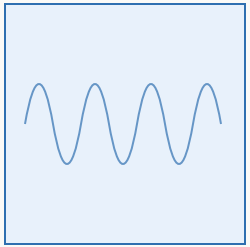
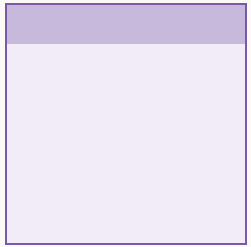


W = window ⊙ e^(–2πi kn/N)

1024 × 513, constant on GPU

=



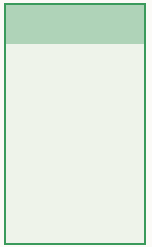
|X|² F × 513

·

mel filters

513 × 128

=



ln(mel)

F × 128

One GEMM kernel serves feature extraction and the Transformer.